Stratigraphic Interpretation of Paleomagnetic Measurements on the Waterberg Red Beds of South Africa

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Samples have been collected from twelve sites at four different localities through the Waterberg red bed succession in the Transvaal. The samples respond equally well to both thermal and magnetic cleaning, and in most cases there is no significant difference between the results obtained by the two methods. After magnetic cleaning at all the sites and correcting for the attitude of the beds in each case, the directions of magnetization form three groups. These groups appear to be related to the stratigraphic position of the sites in the succession. It is suggested that these changes in direction of magnetization through the sequence are the result of considerable polar wandering which took place during the period of deposition and consolidation of the Waterberg sediments. The suggested polar wander path is consistent with the pole positions determined previously from other Precambrian studies in southern Africa. The measurements provide further evidence for the hypothesis that the Waterberg and Umkondo systems should be correlated and that the age of the 'post-Waterberg' igneous activity and the Premier Mine kimberlite is confined within the limits of the Waterberg system. The results also suggest that the Mashonaland dolerites of Rhodesia are slightly older than the Umkondo dolerites, contrary to previous observations, and that their age is also confined within the limits of the Waterberg system.

INTRODUCTION

A previous investigation of the post-Waterberg diabase of the Transvaal and Botswana (Bechuanaland) by *Jones and McElhinny* [1966] has shown that these intrusives can be correlated with the Umkondo dolerites of Rhodesia situated nearly 1000 km to the north. *McElhinny* [1966*a*] has provided paleomagnetic evidence supporting the geological evidence that the Umkondo dolerites were intruded toward the close of the Umkondo system. Consequently, *Jones and McElhinny* [1966] suggested that the Waterberg and Umkondo systems themselves might be correlated. To test this hypothesis a paleomagnetic investigation has been made of the Waterberg red beds.

At one time the Waterberg system was defined to include the rocks that were pre-Karroo in age and rested unconformably on the Bushveld igneous complex and the Pretoria series of the Transvaal system. *Truter* [1949] has suggested that the lower beds of the Waterberg system should be distinguished from the upper beds and proposed that the lower beds be known as the Loskop system. The Bushveld complex has been dated at 1950 ± 50 m.y. [*Nicolaysen et al.*, 1958] and is overlain unconformably by the Loskop system. An angular unconformity separates the Loskop from the Waterberg system, which is intruded by the Leeuwfontein alkaline complex dated at 1420 ± 70 m.y. [*Oosthuyzen*, 1964]. An age of 1790 ± 70 m.y. has also been determined by *Oosthuyzen* [1964] on a sample of granophyre that intrudes rocks that are correlated either with the Waterberg or the Loskop systems. Thus, the Waterberg system may well lie within the limits 1720 and 2000 m.y., and it certainly lies within the range 1350 to 2000 m.y.

The Waterberg system is exposed in five main areas in the Transvaal (Figure 1), the sediments being predominantly red or purple in color. They are largely arenaceous, although conglomerate bands occur usually toward the base of the system. In the Zoutpansberg Mountains the beds attain a thickness of more than 2000 meters and dip at about 20° to the north;

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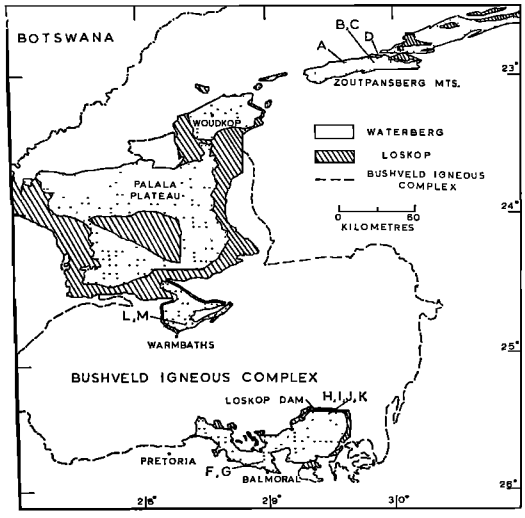


Fig. 1. Map of the Northern Transvaal showing the present exposure of the Waterberg and Loskop systems and the Bushveld igneous complex. The sampling sites are indicated.

numerous faults displace the beds throughout this area. In the Palala plateau, Woudkop and Warmbaths areas (Figure 1), the beds are essentially flat-lying. In the most southerly area the beds are about 2000 meters thick and are nearly flat-lying except on the northern perimeter near the Loksop Dam, where the strata are highly folded with steep southerly dips.

During September 1965, one of us (D. L. J.) visited all the above areas except the Woudkop region and obtained samples from each area visited with the exception of the Palala plateau, where suitable exposures could not be located.

Sampling

Samples were collected from twelve sites in the Waterberg succession in four general localities (Figure 1). Ten of the sites were located in fresh road cuttings. Samples from site F were taken from the side of a hill, and samples from site G were from a stream section. At seven of the sites (A, B, C, H, I, J, and K) the amount and direction of the dip of the beds were measured with a Brunton compass-clinometer; the details of the dip of the beds at the other sites were taken from the geological maps issued by the Geological Survey of South Africa. These data, together with the stratigraphic order of the sites at the four localities, are recorded in Table 1. Samples from site D were taken from sediments that, because of extensive faulting, have no direct connection with the main mass of Waterberg sediments in the Zoutpansberg area. Thus, it is difficult to place this site stratigraphically with respect to sites B and C. A geological section shown on the map issued by the Geological Survey of South Africa suggests the sequence given in Table 1. It is not possible to make an exact correlation of any of the horizons in one area with those in other areas separated by considerable distances across which there is no continuous outcrop. In Table 1, therefore, the vertical thickness sepa-

TABLE 1. Stratigraphic Positions of the Sampling Sites in the Four Localities
Approximate thicknesses of sediments are indicated together with the amount and direction of the dip of the beds at each site.

Zoutpansberg	Warmbaths	Balmoral	Loskop Dam
	Present Upper Limits		
A (29°, 332°E of N) (near top of succession)			>500 meters
		700 meters	H (9°, 191°E of N)
? obscured by faulting			
	???		>300 meters
C (24°, 5°E of N)		F (5°, due north)	I (17°, 170°E of N)
100 meters		80 meters	>300 meters
B (24°, 5°E of N)	L (flat lying)	G (5°, due north)	J (56°, 194°E of N)
?	<200 meters	800 meters	200 meters
D (22°, due north)	M (flat lying)		K (59°, 183°E of N)
~200 meters	<200 meters		~300 meters
	Base of System		

rating the sites in each locality is indicated together with an estimate of the thickness of sediment above and below them. It should be appreciated that these estimates are only very approximate in each case.

The Waterberg red beds have been intruded at various horizons by dolerite sills. The sites selected for the present investigation are, as far as it is possible to tell, at distances in excess of one sill thickness from the sills. There is little chance, therefore, that there has been any remagnetization at the time of intrusion of the sills.

At site A, unweathered blocks were collected and oriented with the Brunton compass-clinometer. At three sites (B, F, and G) cores 60 cm long and 2.5 cm in diameter were obtained by using a modified version of the portable sampling equipment described by *McElhinny and Neal* [1966]. At the remaining eight sites, cores 20 cm long and 2.5 cm in diameter were taken by means of a hand-held petrol driven drill similar to that described by *Doell and Cox* [1964]. At all eleven sites the cores were oriented with respect to the sun's azimuth and the horizontal by means of the method described by *Gough and Opdyke* [1963]. From the oriented blocks 2.5-cm-diameter cores were drilled in the laboratory, and these together with all the other cores were sliced into a number of cylinders 2.5 cm in length.

Measurements

The natural remanent magnetizations (NRM) of the cylindrical specimens were measured with a spinner magnetometer [*Gough*, 1964]. Upon initial measurement, four sites were randomly magnetized according to the criterion of *Watson* [1956], but, in general, the directions at the other sites were usually well grouped away from the direction of the present earth's magnetic field.

Thermal demagnetization characteristics were investigated for one specimen from each of the well grouped sites. Specimens were placed in a nonmagnetic furnace and successively heated to higher temperatures and cooled in field free space (± 10 γ). After each heating and cooling the remanent magnetization was again measured. The normalized thermal demagnetization curves so obtained are illustrated in Figure 2. The behavior of the test specimens fell into three groups similar to the groupings described by *Irving and Opdyke* [1965]. The results from specimens illustrated in Figure 2*a* show that their NRM is made up largely of a *thermally discrete* component of magnetization with a Curie point between 650° and 700°C. Specimens from sites B and C have in addition a *thermally distributed* component with blocking temperatures in the range 400° to 600°C. At 600°C the magnetizations are

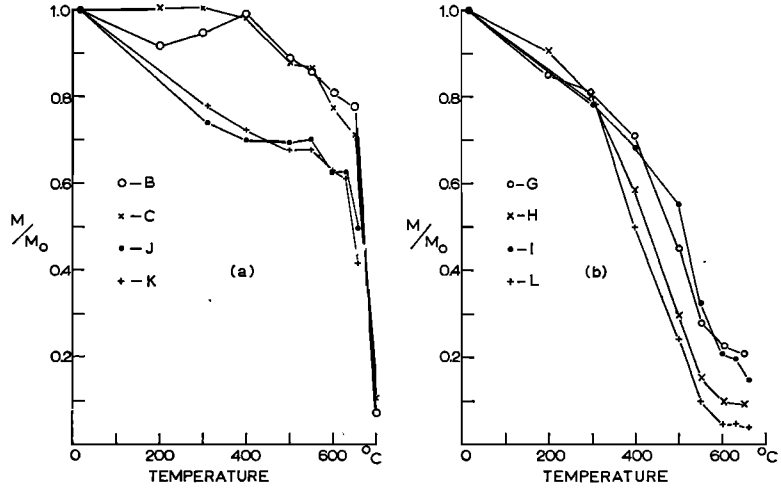


Fig. 2. Thermal demagnetization curves of specimens from the sites indicated. (*a*) Specimens having both *thermally discrete* and *thermally distributed* components; (*b*) specimens having only *thermally distributed* components.

more steeply inclined, and the directions have moved by about 15°. Specimens from sites J and K, however, have a *thermally distributed* component with blocking temperatures between 20° and 400°C. Directions of magnetization of these specimens move through about 50° in this temperature interval and thereafter remain approximately constant up to 650°C. Sites J and K are located in the most steeply folded beds sampled, and the acquisition of a large component with low blocking temperatures may be related to the more severe folding at these sites.

The behavior of the third group of specimens from sites G, H, I, and L is shown in Figure 2*b*, from which it can be seen that the magnetization of these specimens is composed almost entirely of *thermally distributed* components ranging from 20° to 660°C or more. Specimens from each site were also subjected to alternating-field demagnetization in successively higher fields by means of the apparatus described by *McElhinny* [1966*b*]. The normalized demagnetization curves for a specimen from each of four sites are shown in Figure 3. The specimen from site C is very stable under af demagnetization, and, after partial demagnetization in 80 mw m^{-2}, its direction becomes more steeply inclined and shows changes similar to those observed under thermal

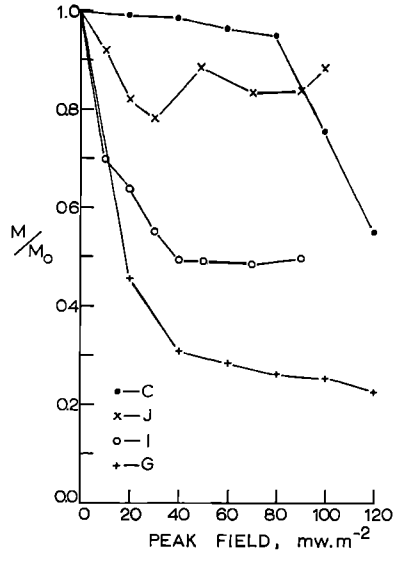


Fig. 3. Alternating-field demagnetization curves of specimens from the sites indicated.

demagnetization. The specimen from site J moves through 60° in fields up to 60 mw m^{-2} and shows a distinct 'end point,' and its changes of direction of magnetization are again very similar to those observed under thermal demagnetization. The specimens from sites C and J, which contain a large *thermally discrete* component of magnetization, have fairly flat af demagnetization curves. Specimens from sites G and I, which are almost entirely composed of *thermally distributed* components, exhibit a fairly steep initial decrease in moment in fields up to 30 mw m^{-2}, but the curves then flatten out and the directions of magnetization reach a distinct end point. End points characterized the directional changes at all sites except site L, where the directions converged in fields between 25 and 30 mw m^{-2} and started to diverge again in higher fields.

Sets of specimens from six sites were subjected to both thermal and magnetic cleaning, the results of which are given in Table 2. At each of five of these sites the NRM site mean direction and the site mean directions obtained after magnetic and thermal cleaning lay in a common plane, and in each case the direction after thermal cleaning lay farther from the NRM direction. Statistically, however, there is no significant difference between the site mean directions after thermal and magnetic cleaning. Also at four of these sites the within-site scatter was greater after thermal cleaning than after magnetic cleaning, and in each of these cases the magnetization was composed largely of *thermally distributed* components. The site mean direction at site L after thermal treatment at 600°C was significantly different from that obtained by magnetic cleaning, but the thermal results are very scattered and may be considered random at the 1% level [*Watson*, 1956]. There may be some significance in the fact that, under magnetic cleaning, no good end points were observed.

In view of the general agreement between the site mean directions after both thermal and magnetic cleaning and the generally higher precision obtained by the magnetic cleaning method, all the remaining sites were subjected to magnetic cleaning only.

The behavior of our specimens under thermal demagnetization follows similar patterns to those observed by *Irving and Opdyke* [1965]

TABLE 2. Site Directions for the Waterberg Red Beds

Site	Site Position		M_n					B_{ac} or	M_{ac} or M_{th}						Site Pole	
	Latitude	Longitude	N	D	I	R	k	Temperature	N	D	I	R	k	α_{95}	Latitude	Longitude
A	22°55′S	29°37′E	4			1.288*		40–160 mw m^{-2}	4	175	+11	3.964	83.7	10	71.8°S	192.8°E
B	22°55′S	29°56′E	4	194	−32	3 845	19	80 mw m^{-2}	4	189	−48	3.810	15.8	24	37 3°S	220 0°E
C	22°55′S	29°56′E	5	178	−30	4 873	31	80 mw m^{-2}	5	176	−41	4.931	57.9	10	43.4°S	204.8°E
D	22°53′S	29°53′E	5			1.395*		10–180 mw m^{-2}	5	145½	−69½	4.645	11.2	24	8 3°S	190.0°E
F	25°47′S	28°59′E	6			3.208*		40 mw m^{-2}	6	119½	−38½	5.928	69.4	8	14.5°S	152.3°E
G	25°47′S	28°59′E	6	144	−60	5.939	81	60 mw m^{-2}	6	110	−39	5.678	15.5	18	7.0°S	147.6°E
								600°C	3	83½	−7½	2 563	4.6	66		
H	25°28′S	29°27′E	5	203	−7	4.010	4	40 mw m^{-2}	5	195½	−3½	4.850	26 7	15	58.8°S	240.3°E
								500°C	5	194	−1	4.196	5.0	38		
I	25°28′S	29°27′E	5	181	−76½	4.916	48	70 mw m^{-2}	5	89	−63½	4 956	91.7	8	18.3°N	161.4°E
								600°C	5	73½	−44½	4.655	11.6	23		
J	25°27′S	29°27′E	5	22½	−9	4.903	41	80 mw m^{-2}	4	13½	+37	3.852	20 2	21	42.3°N	46.3°E
								600°C	6	15½	+43½	5 885	43.4	10		
K	25°27′S	29°27′E	5	14	−20	4.764	17	120 mw m^{-2}	5	14	+39½	4.805	20.6	17	40 3°N	46.5°E
								600°C	5	21½	+52½	4 610	10.2	25		
L	24°50′S	28°21′E	5	89½	−39	4.221	5	30 mw m^{-2}	5	111	−49	4.809	20.9	17	4.4°S	154.3°E
								600°C	5	43½	+23	3.907	3.7	47		
M	24°50′S	28°21′E	5			2 645*		30 mw m^{-2}	5	26	+58½	4.917	48.1	11	21.5°N	49 8°E

N, number of samples measured.
R, length of the resultant of the N unit vectors
D, declination of remanent magnetization.
I, inclination of remanent magnetization.
M_n, NRM
M_{ac}, RM after demagnetization in peak alternating field B_{ac}.
M_{th}, RM after demagnetization in the specified temperature.
$k = (N - 1)/(N - R)$, estimate of precision [*Fisher*, 1953].
α_{95}, semiangle of cone of 95% confidence for mean direction.
* These values refer to the 'resultants' of data which are random on *Watson*'s [1956] criterion.

for the Bloomsburg red beds. However, our specimens, unlike theirs, have responded to magnetic cleaning, and in almost all cases the grouping of the specimens at each site is considerably better after magnetic cleaning than after thermal cleaning. The complete results are given in Table 2, where the site directions quoted are those observed after the strata have been restored to the horizontal by rotation of each bed about its strike.

Discussion of Results

In Figure 4, the site mean directions of magnetization have been plotted both before and after restoring the strata to the horizontal. With the beds in their present attitudes, the directions (Figure 4 *left*) fall into three groups, disregarding the fact that both polarities occur in the groups. Sites J and K are almost reversed with respect to sites A and H in one group; in the second group site M is almost reversed with respect to sites B, C, and D; and sites F, G, I, and L form the third group. There seems to be no obvious explanation for the observed groupings. None of the groups consists only of sites from a single locality, and usually sites within a group bear no stratigraphic relationship to one another (Table 1).

After the strata are restored to the horizontal at each site, the directions of magnetization still fall into three groups, but the constituents of each group have changed. There now seems to be some correlation of the stratigraphy with the various groups. Sites J, K, and M lie near the base of the succession in one group, and sites A and H, near the top of the succession, form a separate group. Sites F, G, L, and I, which are certainly higher horizons than J, K, and M, form a third group. It is possible, therefore, that these results represent a pattern of polar wandering that occurred during the deposition and consolidation of the Waterberg system. The suggested path is shown in Figure 5*a*.

If polar movement occurred on such a scale in this part of the Precambrian, one would expect previously determined pole positions from

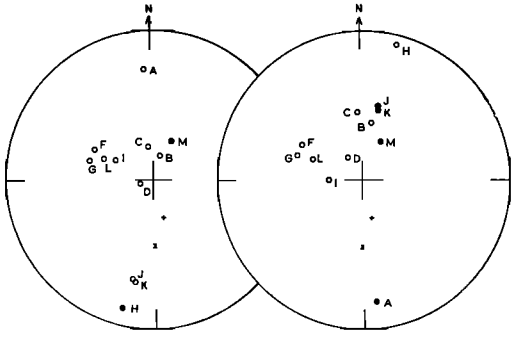


Fig. 4. Lower hemisphere stereographic projection of the directions of magnetization at twelve sites in the Waterberg red beds. The left half shows the directions with the beds in their present position; the right half shows the directions after the beds have been restored to the horizontal. ●—north-seeking direction down. ○—south-seeking direction down. +—present earth's field, south-seeking down. ×—axial dipole field, south-seeking down.

rock units of southern Africa to be consistent with the suggested path in Figure 5*a*. These pole positions are shown in Figure 5*b*, and a possible polar-wander curve, which takes account of the Waterberg results, has been drawn through them. This curve, covering the period from 2000 m.y. to not less than 1350 m.y., is different from that given by *Gough et al.* [1964]. We favor the revised interpretation after consideration of the following points:

1. The Waterberg system is younger than the Bushveld complex, but it is interesting to note that the pole position for the Bushveld gabbro [*Gough and van Niekerk,* 1959] lies close to, and has the same polarity as, the lowest horizons sampled in the Waterberg succession (J, K, and M). It should be remembered, however, that the Loskop system also occurs between the Bushveld complex and the lowest Waterberg horizons.

2. The uppermost beds sampled at sites A and H give poles that lie close to the pole position for the post-Waterberg diabases [*Jones and McElhinny,* 1966], and thus this pole can be expected to be the end of the polar-wandering path covered by the Waterberg succession. Our measurements suggest that the post-Waterberg diabase was intruded during the closing stages of the Waterberg deposition; hence, it is proposed that these should now be referred to as the *Waterberg diabase.*

3. *Jones and McElhinny* [1966] showed that the Waterberg diabase was intruded con-

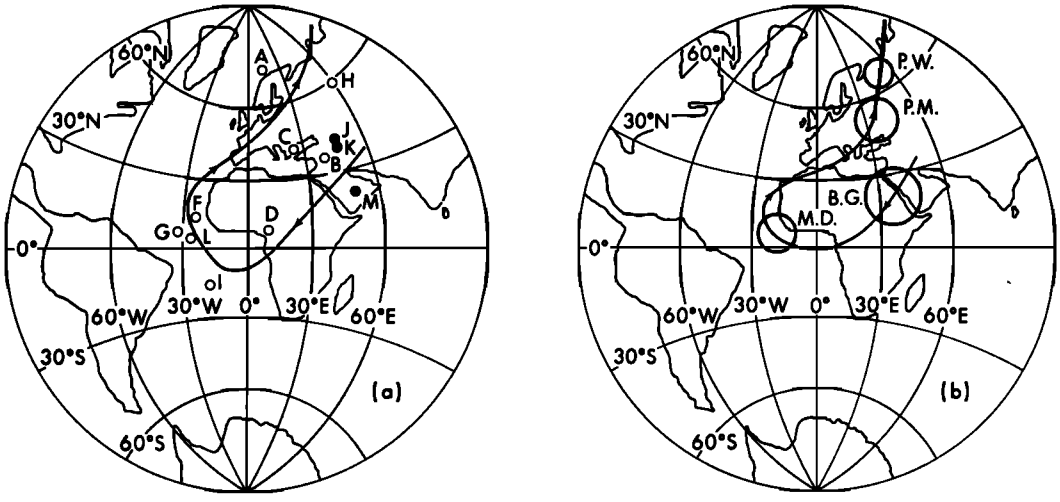


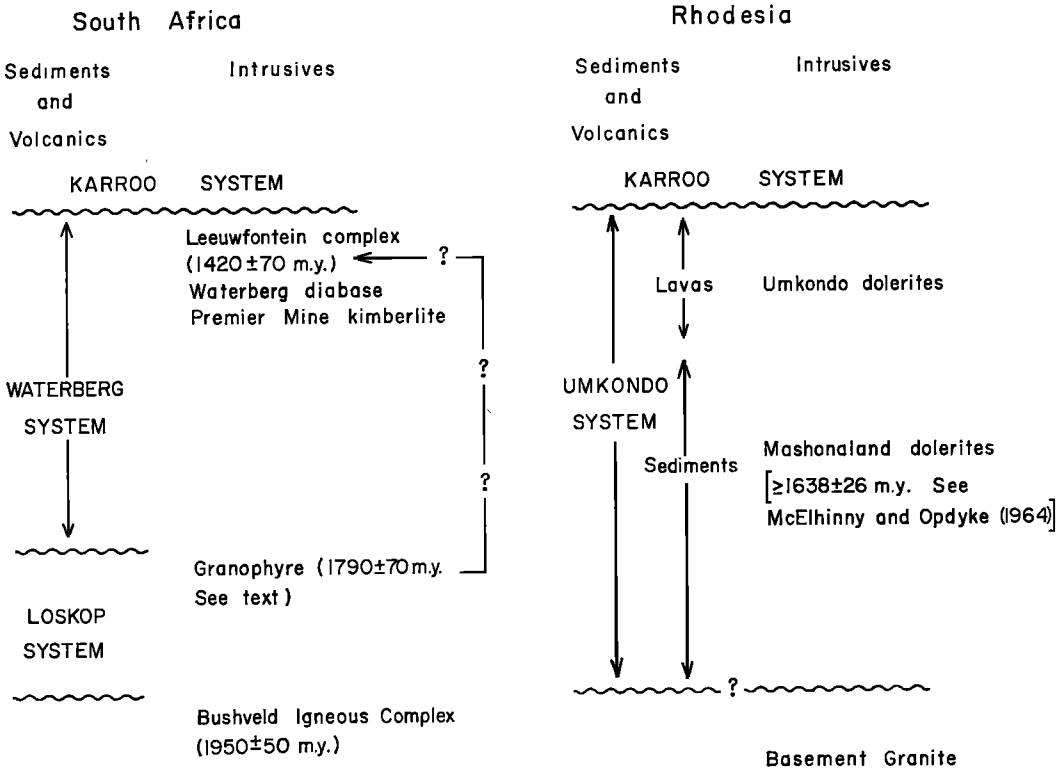
Fig. 5. (*a*) Site poles for the twelve sites in the Waterberg red beds and the stratigraphically inferred polar-wander curve. Solid circles, north poles; open circles, south poles. (*b*) Previously determined pole positions for southern Africa. Circles of 95% confidence are shown for the pole positions of the Bushveld gabbro (B.G.), the post-Waterberg diabase (P.W.), the Premier Mine kimberlite (P.M.) and the Mashonaland dolerite (M.D.) The suggested polar-wander path for this period of the Precambrian is indicated.

temporaneously with the Umkondo dolerite of Rhodesia, which *McElhinny* [1966*a*] has shown to have been intruded in the closing stages of the Umkondo system. Thus, the new curve takes account of the Umkondo dolerite pole [*McElhinny and Opdyke,* 1964], which was not considered in the polar-wander curve of *Gough et al.* [1964] because its age was uncertain at that time. Furthermore, support has been provided for our suggestion that the Umkondo and Waterberg systems should be correlated.

4. It is known [*Jones and McElhinny,* 1966] that the Premier Mine kimberlite is older than the Waterberg diabases and that its pole position (D. L. Jones, 'Study of the paleomagnetism of Premier Mines kimberlites,' in preparation, 1967) is consistent with the new curve. Our measurements strengthen our previously stated hypothesis that the age of the Premier Mine kimberlite is confined within the limits of the Waterberg system.

5. The pole position for the Mashonaland dolerites lies on the new curve in a position that suggests that the Mashonaland dolerites predate the Umkondo (and Waterberg) intrusives, contrary to the relative age suggested by *McElhinny and Opdyke* [1964]. We have re-examined the evidence put forward by these authors for the relative ages of the Mashonaland and Umkondo dolerites. Their argument is based on their observation at site 12 in the Nyhodi valley, Melsetter district, of a direction of magnetization corresponding to the Mashonaland dolerites. This site was located in an outcrop of fine-grained basaltic material occurring in the center of a large sill of Umkondo dolerite, and it was therefore regarded as a separate intrusion into the Umkondo sill. After unsuccessful attempts to find other localities where the Mashonaland dolerite intrudes the Umkondo dolerite [*McElhinny and Jones,* 1965], we have made further studies of the Nyhodi valley and can confirm that the large

TABLE 3. Stratigraphic Correlation of Some Precambrian Systems and Intrusions in South Africa and Rhodesia



sill is Umkondo dolerite. As a result of these studies we feel it is not possible to say with any certainty that the fine-grained basaltic material at site 12 is intrusive into the Umkondo sill, especially as the contact is not exposed. It has been suggested to us as an alternative that site 12 could lie in an exposure of another sill stratigraphically underlying the large Umkondo sill, but brought to the surface at this point through subsequent faulting and erosion. In view of the ambiguity we consider that site 12 does not provide reliable evidence for the relative ages of the Mashonaland and Umkondo dolerites. In view of our results from the Waterberg sediments, we suggest that the Mashonaland dolerites were intruded during the period of deposition and consolidation of the Waterberg system but are older than the Waterberg and Umkondo intrusives. Thus the age difference between the two may not be as great as is suggested by their great difference in pole positions.

The location of the pole from site D on the polar wandering path, as shown, is consistent with its stratigraphic position in Table 1 as are the poles for sites B and C, which are certainly lower in the succession than site A, and have a similar polarity and position to the Premier Mine kimberlite pole.

This interpretation of the results from the Waterberg red beds leads to the Precambrian correlation chart in Table 3. They also indicate that detailed paleomagnetic measurements from all the Waterberg localities in South Africa could be used as a means of correlation through the succession. Such an investigation is planned shortly. A paleomagnetic investigation of the Loskop system is desirable to fill the gap between the Bushveld gabbro and the base of the Waterberg system. The suggested correlation of the Matsap beds, which occur to the west in Griqualand West, with either the Loskop or Waterberg systems [*Haughton*, 1963] could also be tested through a similar paleomagnetic study of these red beds.

The polar-wander curve relative to southern Africa given in Figure 5*b* suggests that movements totaling about 120° took place during the interval between 1350 (or 1720) and 2000 m.y. If the relative ages of Precambrian rocks in other continents could be established, it might be possible to identify a similar loop in their polar-wander curves and, hence, to make intercontinental correlations with southern Africa.

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